# Appendix A

Supplementary material for

*ECDSA.Fail: Open Autoresearch for Optimizing Elliptic-Curve Point Addition in Shor's Algorithm*

*This document reproduces Appendix A of the paper. Section, figure, and table numbers are those of the main paper, so a reference such as* Section 5.2.3 *or* Table 5 *carries the same number in both documents. References to material outside Appendix A resolve in the main paper.*

## A Detailed Results and Circuit-Improvement Trajectory

### A.1 Overall Summary of Results

Under its official scalar objective, the challenge produced a product-score incumbent. Viewed as a two-resource experiment, however, the public archive does not reduce to that single point. It discloses a sparse sample of validator-supported operating points, each instantiating a different answer to the reversible-information question: which values must remain live, and which may instead be recomputed, encoded, measured, or borrowed? For a submitted circuit $c$, let $Q(c)$ denote peak logical width, $T(c)$ the average number of executed Toffoli gates over the challenge validation support, and $S(c) = Q(c)T(c)$ the scalar score defined in Section 3.2. The principal empirical result is therefore an archive-conditioned set of operating points and transitions, not a winning number or a continuously reconstructed resource surface.

Unless stated otherwise, quantitative archive claims in this section use the public-ledger snapshot of 2 July 2026. A six-row source audit completed on 12 July changes the attribution of several mechanisms but not the operating points reported here. This fixed data cutoff explains small differences from later headline records quoted elsewhere in the evolving paper.

Within that snapshot, the first promoted baseline, submission `30c8ded`, used 2,715 peak logical qubits and 3,960,753 average executed Toffoli gates, for a score of 10,753,444,395. The best product-score submission in the same record, `71f5115`, used 1,152 qubits and 1,364,230 average executed Toffoli gates, for a score of 1,571,592,960. Relative to the baseline, peak width fell by 57.6%, average executed Toffoli by 65.6%, and the product objective by 85.4%, a 6.84-fold improvement. The same row is also the lowest-Toffoli promoted circuit in this data cutoff. This conjunction records where the product-oriented search finished; it does not imply that the two objectives are generally aligned.

The lowest-width admitted validator-clean circuit lies on a different branch. Submission `b6f2b0a` reached 825 logical qubits while requiring 489,161,900 average executed Toffoli gates, approximately 359 times the count of `71f5115`. Its product score is correspondingly poor, but the row is scientifically informative:

| Operating point | Submission | $Q$ | $T$ | $QT$ |
|---|---|---|---|---|
| Initial promoted baseline | `30c8ded` | 2,715 | 3,960,753 | 10,753,444,395 |
| Product-score endpoint | `71f5115` | 1,152 | 1,364,230 | 1,571,592,960 |
| Low-$Q$/score bridge | `1c0e0e9` | 1,133 | 1,460,511 | 1,654,758,963 |
| Near-1,000 clean route | `a77c9da` | 1,019 | 31,051,277 | 31,641,251,263 |
| Sub-1,000 Pareto point | `4352cfb` | 980 | 28,847,321 | 28,270,374,580 |
| Sub-950 Pareto point | `a3b0148` | 948 | 54,781,961 | 51,933,299,028 |
| Lowest admitted width | `b6f2b0a` | 825 | 489,161,900 | 403,558,567,500 |

Table 5: Representative operating points in the 2 July data cutoff. $Q$ is peak logical width and $T$ is average executed Toffoli. Rejected rows are used only when admitted under the validation-evidence rule above.

it demonstrates a validator-clean implementation of the point-addition primitive at a width far below the product-score incumbent when transcript, quotient, and route information are compressed into metadata and recomputation. The extreme-width branch thus encounters a steep reversible time–space tradeoff [3,4]. Source reconstruction identifies metadata and denominator-witness lifetime as the immediate constraints within the observed route family.

**A.1.1 Headline operating points** The benchmark evidence requires two qualifications. First, *validator-clean* means that a submission or archived public note records a clean run over the 9,024 Fiat–Shamir-derived validation instances, including the classical, phase, and ancilla channels described in Section 3.3; it is not an all-input theorem. Second, average executed Toffoli is the challenge metric, not necessarily a static emitted-gate count, $T$-depth, or fault-tolerant hardware cost. The archive does not report per-instance dispersion, so these averages are descriptive statistics over the fixed validation support and do not imply a population confidence interval.

For the Pareto construction and related summaries, we call a row *evidence-admitted* if it is promoted or if a rejected submission has a public note reporting a complete validator-clean 9,024-instance run and satisfies the supplementary curation rule. This conservative rule may omit valid circuits whose validation evidence did not survive. The constructions below are therefore conditional on the archive, not exhaustive over every circuit that may have been produced. Hereafter, "admitted" has this evidential meaning.

Table 5 yields two immediate conclusions. The product-score incumbent is not a proxy for minimum admitted width, and the lowest-width admitted artifact is not product-efficient. Their separation records how aggressively recomputation, encoded control, and reversible witnesses can replace resident state before nonlinear work becomes prohibitive.

**A.1.2 Three empirical constructions** The results are most legible through three related but non-equivalent constructions. The first is a historical scalar-incumbent sequence, the second is a mechanism-oriented low-width track, and the third is a cross-sectional Pareto set at the data cutoff. Keeping them distinct avoids using "frontier" for three mathematically different objects.

The *product-score incumbent sequence* comprises promoted submissions that reduced $Q \times T$ when submitted. It culminates at `71f5115` and captures the main trajectory of the public leaderboard. Its largest architectural movements came from decomposing peak width into named owners, folding and hosting inversion transcripts, compressing branch-local control, and releasing carries or predicates at their actual final consumer. This is the construction for which the archive supports its most complete claims about cumulative optimization.

The *validator-supported low-width track* is a mechanism-oriented selection of admitted circuits below the product-efficient regime. It is neither a chronological record sequence nor the set of global minimizers of $Q$. A row enters this track because it is an admitted Pareto point, a source-backed explanatory case, or an admitted neighbouring control used to delimit a reported transition. In ledger order, the resulting q980, q988, q952, q1019, q948, q828, q826, and q825 regimes replace resident arithmetic state to varying degrees with narrower schedules, metadata, and replay. Because this track is selected for explanatory contrast, it supports mechanism comparison rather than frequency estimates over all low-width submissions.

The *admitted* $(Q, T)$ *Pareto set* is the only one of these constructions that is a frontier in the usual multi-objective sense. If $\mathcal{A}$ denotes the admitted rows, then At the data cutoff this set contains seven non-dominated operating points: six rejected low-$Q$ rows supported by full-validation notes admitted under the curation rule, and the promoted product tip. The public branch and submission model retained these artifacts even though a winner-only leaderboard would have discarded them, exposing otherwise hidden trade-offs and resource obstructions.

The trade-offs in Table 2 are highly non-uniform. Moving from 826 to 828 qubits coincides with 31,234,512 fewer average executed Toffoli gates, approximately 15.6 million per added qubit. Moving from 980 to 1,133 qubits coincides with a reduction of more than 27 million Toffoli, whereas the final nineteen-qubit movement from 1,133 to 1,152 coincides with only 96,281 fewer. These secant rates are not causal marginal effects or theoretical lower bounds: the points arise from different route families. They nevertheless provide evidence for a steep hard-$Q$ transition among the observed routes.

**A.1.3 Fixed-budget slices** The archive was not generated by a preregistered fixed-budget experiment: participants optimized different objectives at different times. The following rows are therefore ex post slices rather than controlled

| Budget | Submission | $Q$ | $T$ | Status |
|---|---|---|---|---|
| 1,200 | `71f5115` | 1,152 | 1,364,230 | promoted |
| 1,150 | `1c0e0e9` | 1,133 | 1,460,511 | admitted |
| 1,050 | `4352cfb` | 980 | 28,847,321 | admitted |
| 1,000 | `4352cfb` | 980 | 28,847,321 | admitted |
| 950 | `a3b0148` | 948 | 54,781,961 | admitted |
| 900 | `39a9b5f` | 828 | 448,102,916 | admitted |

Table 6: Minimum admitted average executed-Toffoli cost at or below selected peak-qubit budgets. "Admitted" follows the evidence rule in the main text; only the scalar incumbent was promoted by the challenge.

budget trials. For a peak-width ceiling $q$, each row reports the admitted circuit with minimum $T$ among those satisfying $Q \leq q$.

The identical 1,050- and 1,000-qubit rows in Table 6 are not a transcription error: no admitted circuit between 981 and 1,050 qubits improves on the q980 row's executed count. This repetition is evidence of the archive's discrete and uneven coverage. A controlled extension would choose width ceilings in advance, fix the validator and compute budget, and run the same agent workflow independently at each ceiling.

**A.1.4 External results** Cross-work comparison is meaningful only when the measured object remains visible. Babbush et al. report point-addition operating points in a zero-knowledge proof package. Schrottenloher publishes explicit and tested point-addition architectures, TrailMix supplies open implementation routes, and ECDSA.Fail reports public challenge submissions under its own validator for executed cost [1, 16, 18]. Full ECDLP estimates, distributed per-node estimates, and physical fault-tolerant costs are different objects. Table 3 is therefore a map of neighbouring operating points, not a ranking.

The headline numbers for `71f5115` are lower on both axes than those reported for the 1,175-qubit Babbush et al. point and the open TrailMix `jump-lowqubit` route. This numerical ordering is not an ordering under a common measurement model. ECDSA.Fail counts average executed Toffoli under its validator, some local routes exploit a classical second point, and full window-compatible Shor usage requires the separately validated lookup/use/unlookup construction discussed in the next subsection. The cross-work evidence therefore places ECDSA.Fail in the same broad low-thousands-qubit, low-millions-Toffoli region as neighbouring public results under its own accounting rules. Its distinctive contribution to this comparison is traceability of source history, rejected branches, and local proof obligations, not a cross-harness rank claim.

**A.1.5 Evidential status and selection scope** Claims from the archive vary along two independent dimensions: correctness evidence and benchmark-support selection. On the first dimension, an *exact local transformation* has a value-, quantum-phase-, and ancilla-preserving argument under stated local preconditions. Controlled-swap boundary fusion, exact chunk-boundary clearing, and a borrowed carry lane restored to zero are examples. This supports the local mechanism claim, but it does not prove every parameterization of the enclosing circuit. A *validator-clean circuit*, by contrast, is a complete submitted route with 9,024-instance validation recorded officially or in a public note admitted under the curation rule. This is strong benchmark evidence but remains finite and support-specific.

The second dimension records selection scope. A *support-selected route* may choose truncated comparison windows, width envelopes, q-cap schedules, rerolls, or tail nonces against the benchmark support. Such a route may contain exact local helpers and may be validator-clean as a whole; support selection does not negate either fact. It does block an all-input exactness claim unless a separate proof covers inputs outside the selected support. We consequently reserve exactness claims for source-backed transformations with named local preconditions, describe complete circuits as validator-clean under the stated challenge procedure, and report support selection independently. Formal circuit-family correctness and full-Shor compatibility remain additional proof obligations.

#### A.1.6 Summary interpretation

The central result is not merely that $QT$ fell. The public search repeatedly changed what counted as the scarce object. At first it was an explicit carry or scratch register. It then became a co-owned peak, a resident inversion transcript, a future-owned host location, a branch-window predicate, a denominator witness, and finally a single carry whose lifetime had been conservatively over-approximated. The low-$Q$ branch completes the picture: when increasingly many resident values are converted into metadata or recomputation, width continues to fall but Toffoli cost rises by orders of magnitude. Together, the incumbent sequence, low-width track, and admitted Pareto set provide an empirical anatomy of how reversible information was represented and transformed within this challenge's optimization lineage for `secp256k1` point addition.

### A.2 Detailed Circuit-Improvement Chronology

The trajectory is best read as a sequence of changing resource models. A narrative based only on the leaderboard would present hundreds of descending scores and suggest steady progress. The public record shows a more structured process: long periods of nonce search and local shaving were interrupted by commits that changed the representation of the circuit's live information. Those commits did not merely improve the current design; by altering the location or lifetime of the peak, they made different optimizations technically accessible and newly valuable.

The archived dataset contains 780 public submissions: 397 promoted, 229 rejected, and 154 failed. A first-pass heuristic classifier assigns the 397 promoted rows to 126 nonce or micro shavings, 112 local Toffoli optimizations, 60 structural optimizations, 51 local width shavings, 28 regime transitions, 18 architecture or regime shifts, one structural width cut, and one baseline. These categories are editorial rather than intrinsic properties of the commits, and no inter-rater agreement statistic is claimed. They characterize the curation pipeline rather than observer-invariant natural classes. The imbalance is nevertheless informative: most promoted movement was exploitative, whereas a much smaller set of architectural episodes changed what subsequent search could productively optimize.

The three eras below are an ex post analytical periodization, not stages fixed before the search. We use causal language in a process-tracing sense over software artifacts [8,10]: a later optimization is described as enabled by an earlier one only when temporal order, source changes, route notes, and the later mechanism's preconditions form a coherent dependency chain. Temporal succession alone is not causation. The descriptive unit is the submission row; the unit of mechanism-level interpretation is the *mechanism episode*: an enabling source change, its proof obligations, the operating-point movement, and nearby controls or failures that delimit the claim. This supports a bounded dependency interpretation, not experimental identification of the episode's isolated effect.

**A.2.1 Optimization of the logical width–work product** The union-bound interpretation of the longitudinal proxy requires an explicit scope condition. If $e$ were the applicable per-addition error probability, the union bound would give $1-28e$ as a lower bound on success. The plotted $\hat{p}'_{\mathrm{LB}}$ instead substitutes the empirical $1-\hat{p}$ for $e$: it is a descriptive plug-in proxy, not a confidence bound, and $\hat{p}$ measures classical-output agreement rather than coherent circuit success. Only under an independently rerunnable success model would a 5% composed error imply the repetition factor $1/(1-0.05) \simeq 1.053$, or approximately 5.3% additional expected work.

Throughout this subsection, $QT$ denotes the challenge's product of logical width and nonlinear work. Although sometimes called a spacetime surrogate, it is not circuit depth, active volume, or fault-tolerant spacetime.

*From monolithic arithmetic to jointly owned peaks.* The baseline exposed the familiar costs of reversible modular arithmetic: carry lanes, reduction scratch, controlled swaps, and Euclidean inversion history [5,15]. Early optimization treated these as independent targets, a view that failed whenever several phases tied for the global maximum. Reducing one owner simply revealed another at the same height.

The `d35d6e7` episode fused consecutive Kaliski $(r, s)$ controlled swaps across an iteration boundary (Section B.6). Its immediate width effect was modest, but it established that a control value need not be materialized twice when its forward and reverse lifetimes can be joined exactly.

The first large width movement, `f94f726`, reduced the peak to 2,310 qubits. Late-iteration clean carries, future-pool borrowing, low-scratch Solinas reduction, affine multiplication narrowing, and controlled-swap framing had to move together. The source-backed lesson is that the global peak is a property of overlapping owner intervals, not of the largest helper in isolation.

Once this transition exposed a stable 2,310-qubit regime, the earlier fusion idea could be extended. `112d5a4` applied the same merge to the $(u, v_w)$ channel over its provably safe iteration prefix, reducing average executed Toffoli to 3,361,759 without changing the peak. Subsequent fixed-width exploitation lowered the count to 2,462,914 at `375fdff`. The sequence matters: a boundary-fusion experiment preceded the broad width transition; that transition then created a stable regime in which stronger fusion and local optimization became valuable. Era I reduced the score from 10.75 billion to 5.69 billion, but left a question that local arithmetic tuning could not answer: why was so much inversion history still resident at the peak?

*From resident history to phase-local transcript.* Here "phase-local" denotes a scheduled arithmetic phase; quantum phase and phase-correction obligations are named separately where they arise. Submission `0620443` exploited the changing shape of the Euclidean walk. Dialog-history entries were folded into idle high bits of shrinking registers, carry pools were relocated onto clean suffixes, and an affine square was recomputed instead of carried. Width fell from 2,309 to 2,025 qubits even as Toffoli rose slightly. Bennett-style recomputation and Luo-style register sharing supplied the conceptual vocabulary [2, 3, 12]; the local contribution was an ownership map precise enough to survive reverse replay.

That ownership map enabled `0c1d4d9` to absorb a compressed-sidecar architecture and reduce the circuit to 1,698 qubits. This was not the invention of transcript compression or of the imported point-addition family. Its contribution was integration: compressed-block primitives, lifecycle replay, terminal-state handling, route defaults, and point-addition wiring were made mutually valid under the challenge. Once explicit history ceased to be the dominant object, the 1,698-qubit band could be mined for Toffoli reductions, falling from 2,447,846 to 1,704,086 average executed Toffoli before width moved again.

Submission `744b274` converted the future transcript from passive storage into a host. Future-owned slots carried gated registers before their eventual owner became live, while windowed apply carries prevented the arithmetic from recreating the peak. This reduced width to 1,572 qubits and changed the design question: after broad transcript compression, how little branch and window information did the current arithmetic phase require?

Source inspection reconstructed six transitions whose attribution was initially open, sharpening the cumulative sequence. In `3bd3aa5`, the circuit materialized $f = ctrl \wedge a$ one chunk at a time, cleared each slice with the benchmark's HMR X-basis demolition measurement, and removed retained boundaries with exact controlled comparators. The resulting row used 1,567 qubits and 1,689,505 average executed Toffoli. Submission `da05eeb` then relieved six co-binders together: it hosted the $z_0$ square carry lane, selected a low-$Q$ $z_2$ square, vented Solinas corrections, and widened the exact apply cut, producing a 42-qubit reduction and a 1,500-qubit operating point. With that peak displaced, `f2cd713` hosted a fused branch comparator's carry lane on a clean future-log slice and sank the apply phase to the next body floor, reaching 1,466 qubits.

Submission `ca1cd1f` next applied phase-local reasoning to the transcript through a high-$u$ compressed-log runway and $K = 2$ bounded-shift storage. The circuit remembered only the branch/window information needed to reverse the current arithmetic phase, reaching 1,394 qubits. The final reconstructed transition, `67be862`, combined a low-$Q$ final ripple, a hosted split-boundary comparator, and five balanced chunks to reach 1,320 qubits. These reconstructed rows do not each become an independent discovery. Rather, they show that the earlier "source gap" label denoted an incomplete causal evidence chain, not an inexplicable commit.

Submission `4b7d727` then exploited a different distinction: emitted work and executed work are not the same benchmark object. Conditional replay reduced average executed Toffoli to 1,404,058 at 1,300 qubits by performing nonlinear work only on branches that required it. This is a source-backed challenge-score mechanism, but not automatically a static gate deletion or physical-resource reduction. By the end of Era II the score had fallen to 1.83 billion. The broad architectural objects had been compressed; what remained were increasingly small, proof-sensitive lifetimes.

*From broad compression to last-consumer proofs.* The descent through the 1,226–1,168-qubit region was jagged. Route defaults, window widths, support assumptions, rerolls, and unrelated co-binders could decide whether a one-qubit change appeared at the measured peak. Many rows in this band remain evidence-curation targets, so their mechanisms are left unattributed rather than reconstructed by inference. Collectively, they show that the circuit had become a packing problem over branch-local transcript state, shrunken arithmetic, and narrow validation islands.

The 1,165–1,169 regime persisted for approximately 90.7 hours and 113 public submissions before `44d7d51` entered the 1,160–1,164 bucket. The unblock was a one-bit lifecycle argument. A bit known to be zero or one at a particular phase could host temporary information provided its original owner was dormant and the bit was restored before that owner returned. The general idea is old; the local scientific work was to name the exact value, owner interval, and restoration point at a hard-pinned peak.

Submission `6c1a65d` generalized this reasoning from a bit to a schedule. Per-call reserve maps replaced route-global scratch assumptions, final carries were consumed directly where legal, and measured cleanup was paired with the nonlinear phase feedback required by the branch relation [6, 9]. Finally, `71f5115` released the `cy0` carry in the internally named FFG arithmetic schedule on suffix calls where its last consumer could be named, while coordinated square-vent, coordinate-truncation, reverse-subtract, and tail-swap changes prevented adjacent phases from reclaiming the maximum.

The late product-score sequence therefore closes with a precise proposition rather than a grand identity: this carry is dead, on these calls, after this consumer, under this route. Progress came from replacing conservative residency with increasingly exact statements about who owns information and when.

**A.2.2 Low-qubit track** The low-qubit track asks how far width can be reduced when Toffoli cost is not the primary objective. Reversible computation permits time to be exchanged for space, but the exchange rate is architecture-specific [3, 4]. The public rejected branches make that exchange empirically visible for the route families under study.

TrailMix supplied the principal inbound implementation substrate. Its open `shrunken-PZ` route reported approximately 1,050 qubits and 32.3 million Toffoli, demonstrating a qualitatively different point-addition architecture below the main low-thousand-qubit routes [18]. ECDSA.Fail did not invent that architecture; it provided a public environment in which the route could be ported, specialized, stress-tested, and compared with neighbouring circuits under a common validator.

*Sub-*1,000 *specialization and coordinate liveness.* Submission `cb40049`, source `86c8e1e`, combined three changes: a deterministically trained thin schedule followed by a 500,000-sample repair pass, an 8-qubit completion counter, and direct classical-register modular additions and subtractions that avoided a 257-qubit loaded-coordinate register in the highest-pressure coordinate phases. The profile retained the baseline leading-zero scan and 6-bit shift-distance control, and selected tail nonce 2. Its attached public note records a clean local 9,024-instance run at $Q = 999$ and $T = 57{,}041{,}103$. The service row is failed and has no API metrics, so these figures are local-note evidence rather than service-reported resource metrics.

Submission `85d5dae`, source `f46a8c4`, then introduced an 80-bit leading-zero scan window, capped the quotient at 20 qubits, and reduced the shift-distance register to 5 qubits while retaining the 8-bit counter. The source regenerated the thin schedule with seed 278 and selected tail nonce 602. Its API row reports $(Q, T) = (980, 39{,}648{,}420)$, and its public note records a clean local 9,024-instance run. These settings form a joint profile: the observed reduction is not attributable to any one bound or nonce in isolation.

The lower-work branch changed the lifetime of the forward numerator. Source `6020c67` reversed $\Delta y = \lambda \Delta x$ immediately after the final consumer of $\Delta y$, reused the resulting clean register for loaded-coordinate arithmetic, and rematerialized $\Delta y' = \lambda \Delta x'$ before output-side slope cancellation. Combined with a 78-bit leading-zero window, step-specific shift bounds with a one-bit safety margin, and tail nonce 3116, this route yielded submission `e0281a8` at $(Q, T) = (988, 29{,}210{,}249)$. Submission `4352cfb`, source `32bd568`, combined the same zero-$\Delta y$ route and 78-bit window with the 20-qubit quotient cap, 5-qubit shift-distance register, deferred $y$-materialization, and tail nonce 25, yielding $(980, 28{,}847{,}321)$. Its average executed Toffoli count is 27.2% below that of the earlier $Q = 980$ profile, but this remains a profile-level comparison because several settings changed together. The API reports both operating points, and their attached public notes record clean 9,024-instance runs.

The subsequent chronology remained jagged. On 19 June, `0ce9b13` moved to 952 qubits while raising the executed count to 78,159,965.

Later that day, the search returned to a wider 1,019-qubit construction. Submission `a77c9da` combined support scheduling, leading-zero windows, margin and shift controls, route defaults, and diagnostic high-bit checks in a validator-clean circuit at 31,051,277 average executed Toffoli. Its importance is explanatory rather than Pareto-optimal: the source record exposed the central obstruction. A second `shrunken-PZ` cancellation suggested an algebraic batch relation of the kind used to amortize simultaneous inversions [7, 14], but a reversible implementation still needed the denominator, cofactor, inverse, and cleanup witness to have a peak-safe owner. Carrying a full-width passenger through inversion erased the width saving. Algebraic feasibility therefore failed to imply circuit feasibility.

On 21 June, `a3b0148` reached 948 qubits at 54,781,961 Toffoli, improving both coordinates relative to q952 but not restoring the low executed cost of q980. The sequence is non-monotonic in both resources because support schedules, arithmetic helpers, and metadata representations change discretely; not every integer width has a comparably efficient circuit.

The hard-$Q$ continuation made metadata the dominant resource. Quotient lengths, implicit high bits, route classes, hosted predicates, and reversal witnesses were encoded instead of storing the full values they described. Submission `39a9b5f` reached 828 qubits at 448,102,916 average executed Toffoli; `f1d8707` reached 826 at 479,337,428; and `b6f2b0a` reached 825 at 489,161,900. The final few qubits thus cost tens of millions of additional Toffoli. Together, these points delimit the observed hard-$Q$ cliff.

This track makes two contributions. Technically, it identifies denominator-witness and metadata lifetime as limiting objects within the observed routes after ordinary arithmetic scratch has been compressed. Methodologically, it shows why rejected submissions belong in the research record: none improved the product score, but together they reveal a resource trade-off that the product leaderboard cannot express.

**A.2.3 Pareto-frontier exploration** The Pareto status is a cross-sectional property of the admitted set at the data cutoff. Historical promotion asks a different question: did a row improve the scalar score when submitted? The two classifications can disagree, particularly on a discrete, non-convex frontier where scalarization conceals useful operating points. Only one of the seven admitted Pareto points is the promoted product tip. The other six are rejected low-$Q$ rows supported by full-validation notes admitted under the curation rule.

The discreteness of the mechanisms explains why both the Pareto set and the chronological search path are jagged. A new host becomes available only after its future owner is proved inactive; a carry disappears only after its final consumer is moved; a transcript shrinks only after a replay relation changes; and a denominator batch helps only if its witness can cross the inversion peak cheaply. These transformations create islands and cliffs rather than a smooth family indexed by qubit count.

This observation makes the open-autoresearch argument empirically specific. A greedy scalar leaderboard supplies exploitation pressure but can suppress branches that spend one resource to explore another [13, 17]. The public submission ledger partially mitigated that loss by preserving rejected commits and notes. A future challenge could make the repair explicit by promoting the $(Q, T)$ Pareto set, maintaining objective-specific leaderboards, or rewarding source-backed regime transitions that do not improve the current scalar tip. Such a design would reward validated information about the geometry of the search space, not arbitrary regressions.

#### A.2.4 Live-information model

The mechanism episodes can be compared through the explanatory tuple

$$\mathcal{I} = (\mathsf{value}, \mathsf{width}, \mathsf{location}, \mathsf{owner}, \mathsf{live\ interval}, \mathsf{provenance}, \mathsf{cleanup}, \mathsf{proof\ duty}). \tag{1}$$

This is a coding schema distilled from the source record, not a circuit theorem or a claim of priority over reversible pebbling, register sharing, or measurement-based uncomputation [2, 3, 5, 6, 9, 11, 12]. It makes explicit two conditions that scalar deltas conceal: a local transformation changes $Q$ only if its live interval intersects a measured maximum and every phase tied at that maximum is also relieved; holding the validation support fixed, it changes average $T$ only if it reduces executed Toffoli work on that support. Transcript compression changes width and provenance; hosting changes location and temporary custody; loans shorten live intervals; measurement changes cleanup and its phase obligation; and hard-$Q$ metadata replaces a full value with a narrower description whose reversal is more expensive. Complete field definitions and evidence requirements appear in Table 7.

The trajectory is therefore an interacting sequence rather than a collection of independent tricks. Joint-peak analysis made transcript residency visible; compression made branch-local provenance salient; measured cleanup made phase part of the resource contract; and low-width exploration made witness lifetime salient. The schema organizes both the 6.84 score ratio and the orders-of-magnitude work increase on the extreme-width branch, but neither decomposes those outcomes causally nor claims predictive completeness. The next subsection changes the analytical axis from chronology to transformation family.

#### A.2.5 What the trajectory shows

Across all three tracks, the underlying research object is live information. As an analytical abstraction distilled from the source-backed episodes above, a local live object may be represented by the tuple in Equation (1). This tuple is an explanatory coding schema, not a circuit theorem or a canonical representation.

Its ingredients are individually established. Reversible simulation and pebbling supply store–recompute trade-offs; reversible addition and register sharing

| Field | Meaning | Evidence required |
|---|---|---|
| Value | Logical datum at the relevant program point. | Forward semantics and applicable range or branch condition. |
| Width | Logical qubits occupied at the measured peak. | Allocation or peak trace tied to the active route. |
| Location | Physical register or slice carrying the datum. | Allocation trace and any host or alias relation at the relevant phase. |
| Owner | Logical object whose future correctness constrains reuse. | Non-interference during borrowing and restoration before ownership returns. |
| Live interval | Production or loan entry through final consumption or restoration. | Producer, last consumer, and reverse-path evidence. |
| Provenance | Stored, recomputed, metadata-decoded, measured, or borrowed. | Source path establishing the current representation. |
| Cleanup | Reversal, consumption, measurement with feedback, or exact overwrite. | Value, ancilla, and phase restoration under stated preconditions. |
| Proof duty | Condition licensing the transformation. | Local proof, exhaustive helper test, or bounded validator evidence. |

Table 7: The live-information coding schema used to compare source-backed mechanism episodes.

supply carry-lifetime and ownership patterns; and measurement-based uncomputation supplies a distinct cleanup relation [2,3,5,6,9,11,12]. The contribution claimed here is neither a new reversible-computation theorem nor priority over those techniques. It is their empirical synthesis into a source-auditable scheme that binds each resource transformation to a live object, cleanup route, and proof duty, permitting comparison across heterogeneous commits.

The schema describes local objects, whereas $Q$ and $T$ are global circuit measurements. The connection is conditional. A local width reduction changes $Q$ only when its live interval intersects a measured maximum and every phase tied at that maximum is also relieved. Holding the validation support fixed, a local work reduction changes $T$ only when it reduces executed Toffoli work on that support. An exact local transformation can therefore be a genuine mechanism without moving either headline metric by itself; joint-pin relief and conditional replay are the clearest examples.

Each source-backed mechanism changes at least one field of Equation (1). Transcript compression changes width and provenance; hosting changes location and temporary custody; a lifecycle loan shortens the live interval; recomputation replaces residency with a cleanup path; measurement-based uncomputation changes cleanup and its phase proof duty; and hard-$Q$ metadata replaces a full value with a narrower description whose reversal is more expensive. The schema is not claimed as a complete ontology. A material source-backed transition that cannot be represented without erasing an important technical distinction would require it to be extended. Nonce and parameter-only rows are search outcomes rather than contrary mechanism cases.

The order of discovery also becomes legible through this schema. Early in the challenge, values and allocated locations were visible, while logical owner and live interval remained implicit. Joint-pin analysis made ownership explicit; transcript compression made provenance explicit; measured cleanup made quantum phase part of the resource contract; low-$Q$ exploration made witness lifetime explicit; and late carry release reduced progress to the last consumer of one bit. The trajectory is therefore not a collection of isolated tricks, but a progressive refinement of the circuit's information model.

**A.2.6 The Circuit-Improvement Search as a Damped Newton Descent.** It is tempting to read the open autoresearch effort for circuit improvement as discussed above as a greedy descent, in which each submission simply steps in whatever direction most reduces the objective. For an easy-to-evaluate, hard-to-solve problem, this reading is accurate only for the dense stretches of small local optimizations, and it is silent about the larger unblocks where most of the insight lives. The reason is that the feasible set of valid solutions is usually non-smooth and laced with constraints. In the ECDSA.Fail case a single shared register can make a genuine reduction in width invisible, a residual phase can invalidate a solution that is otherwise numerically correct, and an algebraic shortcut can be worthless if the quantity it saves must remain live through the most expensive region. On such a surface the direction in which the score last improved, the first-order information, is not enough to decide where to go next.

A more faithful description is that the loop behaves, in a loose but useful sense, like a damped Newton method. A participant proposes a step, measures the residual with the evaluator and the score, inspects the source and the accumulated notes to learn which local constraint was actually binding or which apparent direction was in fact flat, and revises a working model of the frontier before stepping again. The relevant curvature is not a numerical Hessian but the discovered structure of the problem: in this challenge, which live quantities jointly own the peak, which intermediate windows must survive, and which cleanup obligations a given route commits to. The damping is supplied by the exact evaluator and the correctness contract, which hold back an attractive but unproven step, such as a gate cut that lowers the count or a slick algebraic identity, until its value, phase, and ancilla behavior have survived verification. Verification thus plays the role that a trust region plays in optimization, keeping the search from taking a confident step onto an infeasible point.

The consequence that generalizes beyond this problem is that failed proposals are not wasted searches. A rejected port, a candidate that a closer look reveals to hide a resource deficit elsewhere, or a note establishing that an apparent local saving was really a global side effect, each carries second-order information about the local geometry of the feasible set. Such results identify which directions are flat, which cross a constraint surface, and which are worth re-parameterizing, and they are exactly what make the accepted steps intelligible. An open autoresearch record that logs only its successes is therefore a curve without an explanation. Preserving negative results as first-class, shared artifacts is what lets the community build an increasingly accurate model of a hard feasible set, and it is a large part of why the collective search converges faster than any single greedy trajectory could.

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